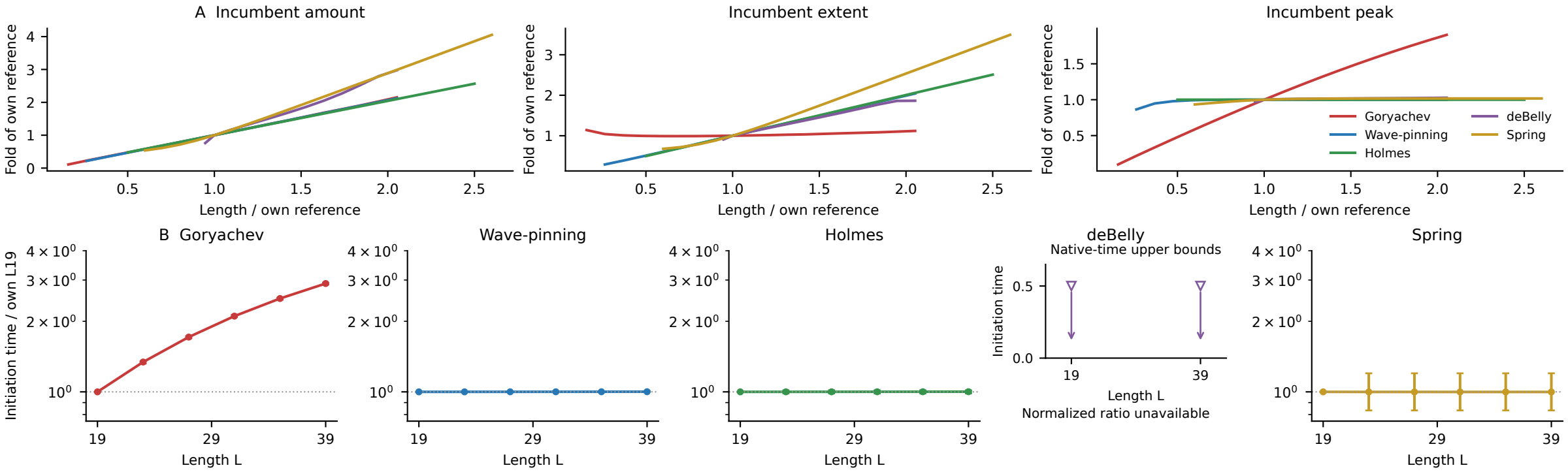
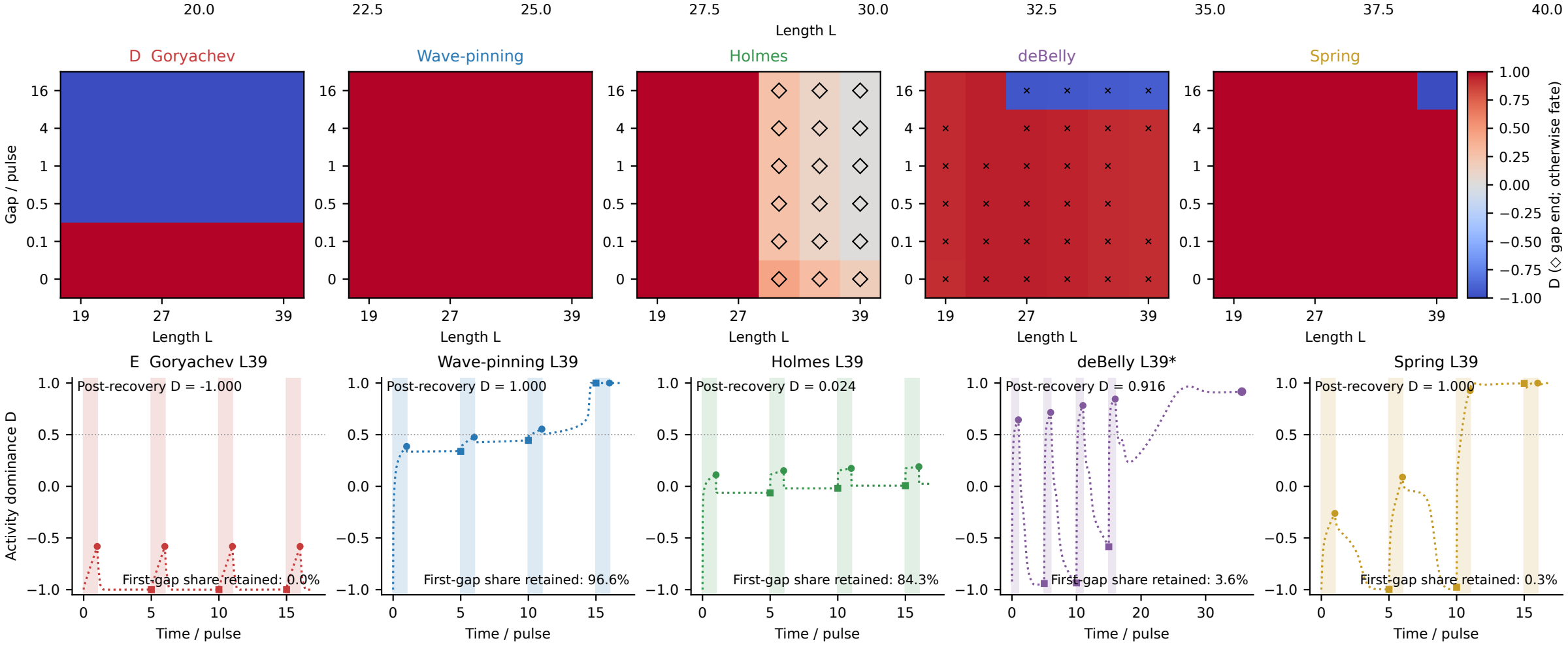
Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures





F Shape and timing signatures distinguish mechanism classes

Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Experimental measurement
Initiation-limited Goryachev	Yes (2.90×)	Modest (1.12×)	Resets	No	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Mass-conserved, saturating Wave-pinning, Holmes	No resolved slowing	Large; grows over resolved L range	Share retained (ratchet)	—	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Slow non-activity state deBelly*, Spring	Unresolved* / no	Modest (~1×)	Collapses early	Yes	Beam width at fixed photon dose; pulse trains; gap-end biosensor share

◇ Holmes L31–L39: last inter-pulse gap-end D, with no target-length calibration. × failed field tail; ? uninformative ratio.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

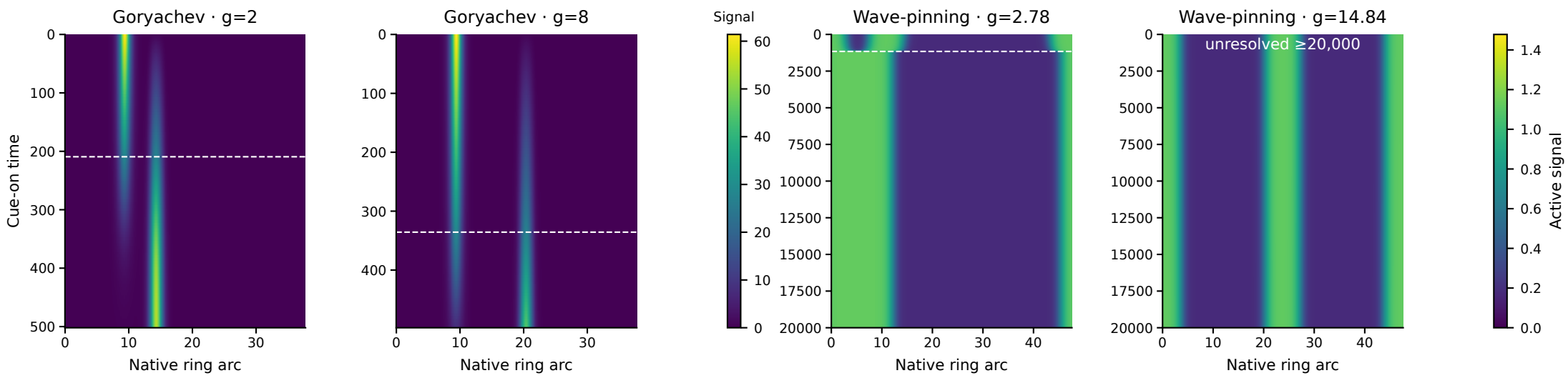
A Rotating a cue separates distance from native cell size



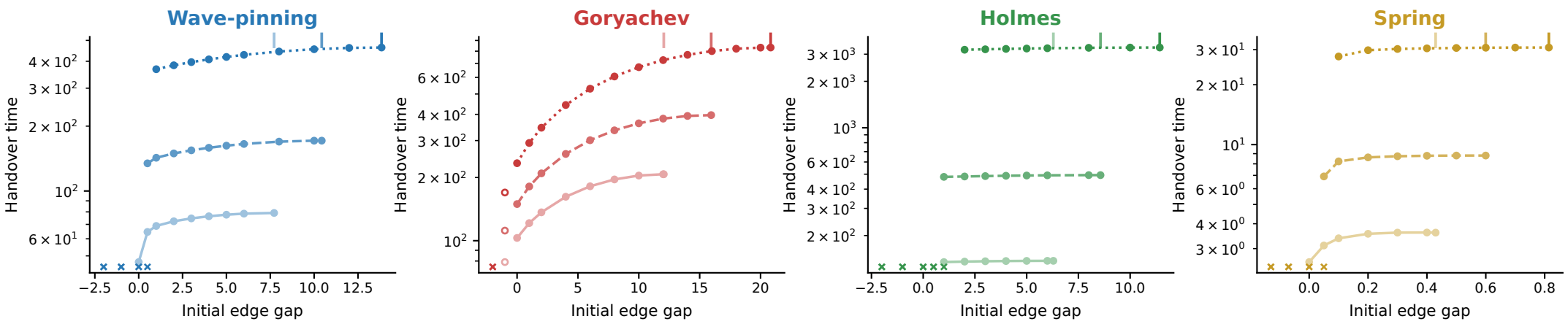
Same physical gap across sizes

A1: continuous cue → handover
B: equal complete-region amounts,
cue withdrawal → single-peak resolution
Blue incumbent; orange cue; g is edge gap.

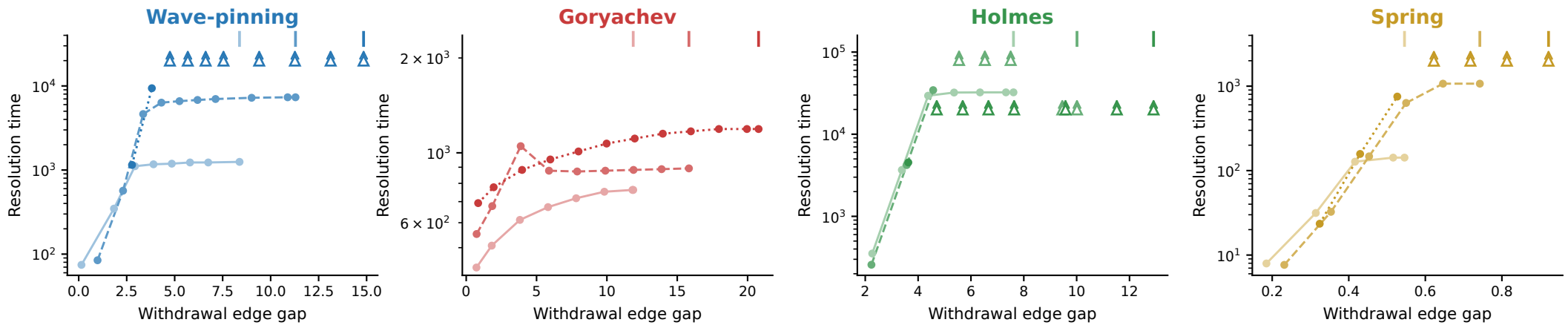
B Native histories: driven handover (left); time after withdrawal (right)



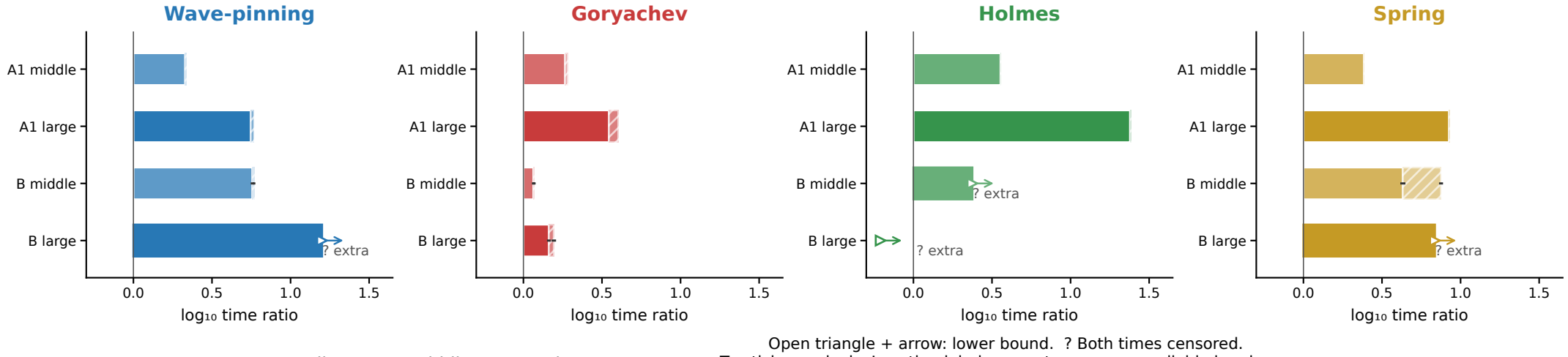
C Driven handover against initial edge gap



D Cue-free resolution against measured withdrawal gap · equal amounts



E Size at matched gap (solid) + extra separation (hatched)

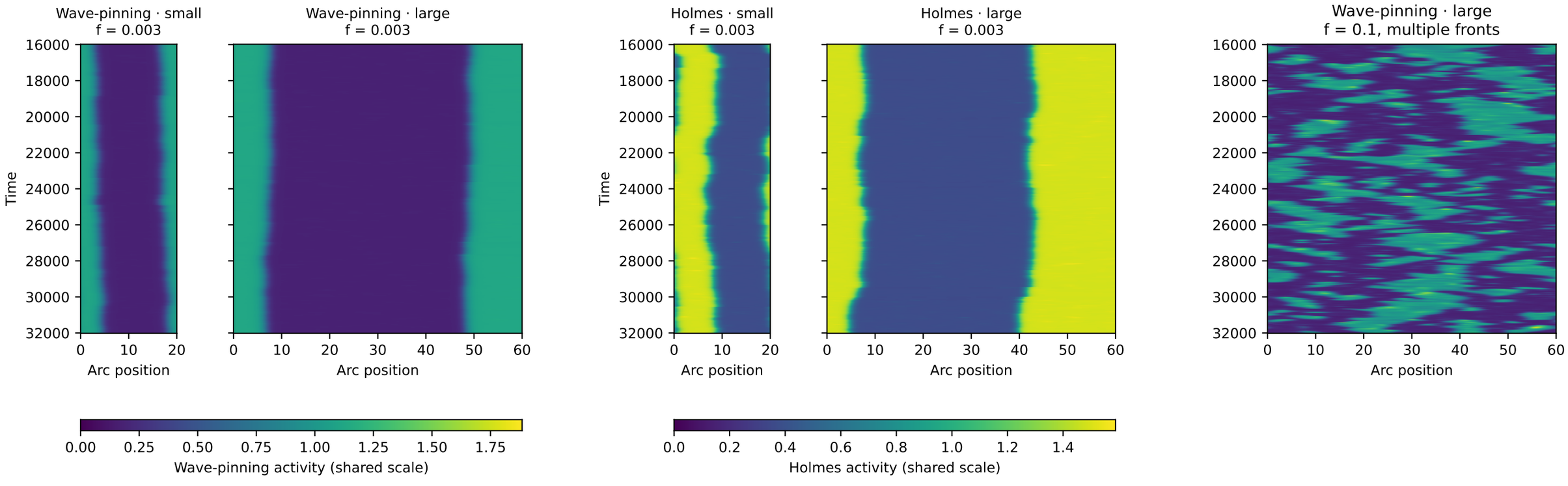


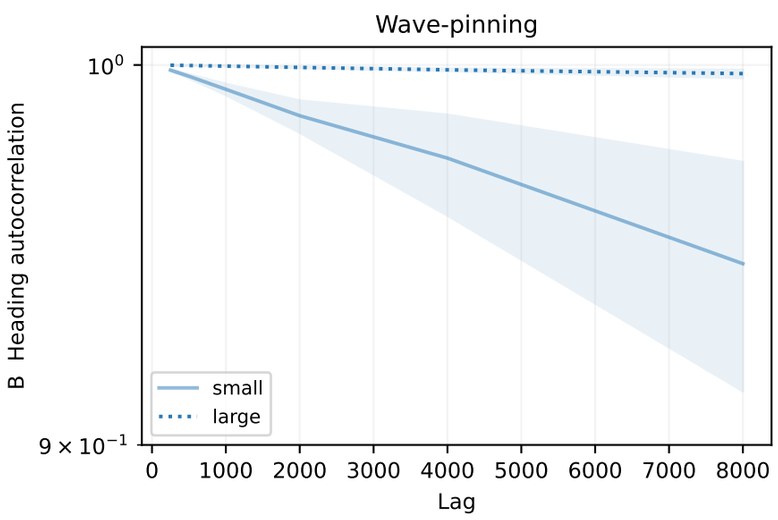
Open triangle + arrow: lower bound. ? Both times censored.
Top ticks: each size's antipodal-placement gap. ×: unavailable handover.

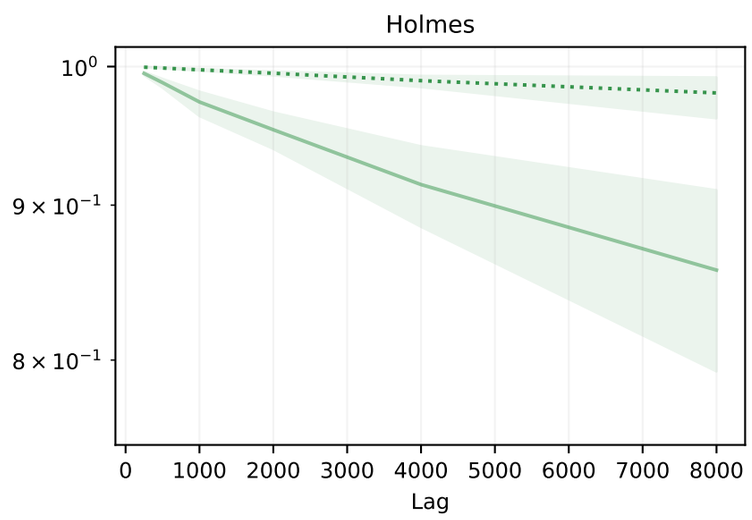
Enlarged cells hold heading longer under fluctuating input

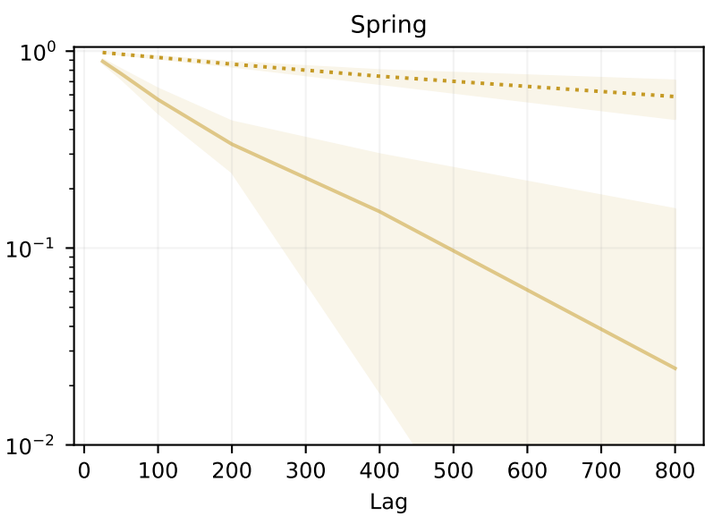
Gains reach or exceed geometry at each model's weakest measurable input; the advantage declines as input strengthens

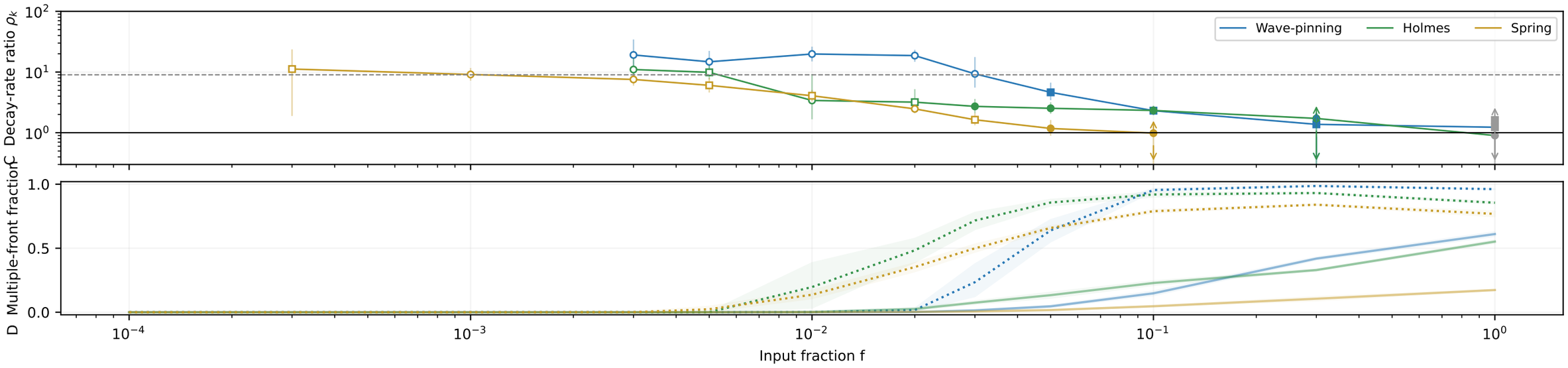
A Native activity histories

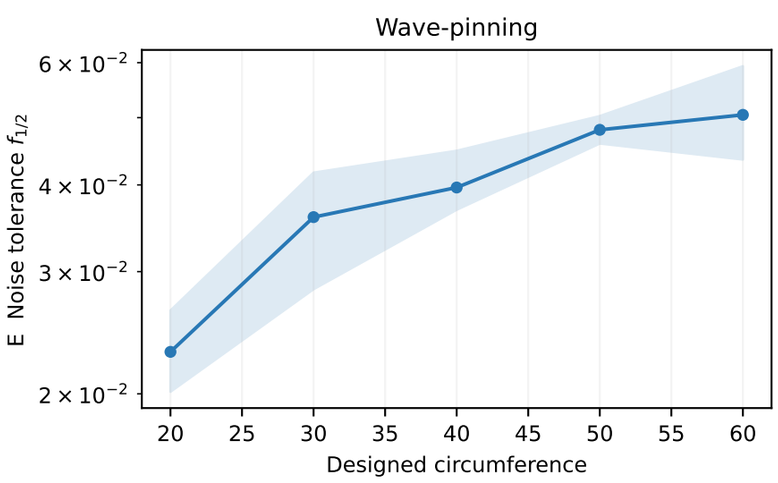


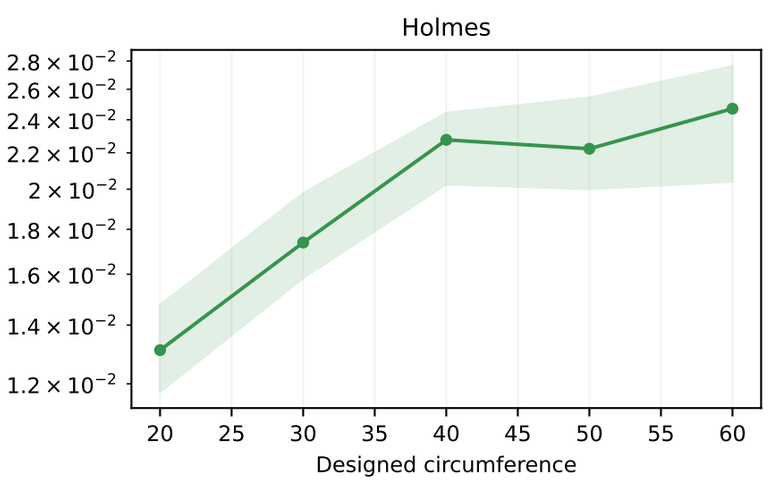


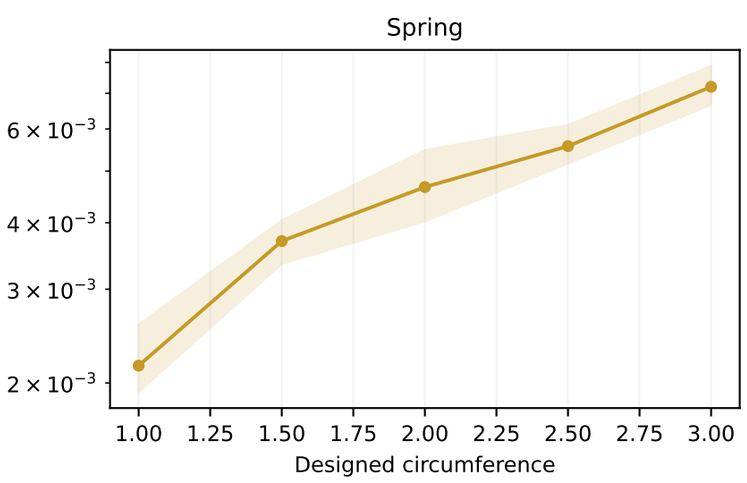












Small: lighter solid; large: darker dotted. C: open fill single front; filled multiple fronts; grey fill low contrast; square >1.5× lag drift.
Pointwise whole-seed 95% CIs. Horizontal references: ratio 1 and nominal geometry 9. D is descriptive; it does not establish a cause of erosion.
H1 uses a post hoc f < large-cell f½ domain; frozen H1 remains inconclusive. Goryachev retained in supplement only.